Assignment 1

PCCST501 – Computer Networks

Submitted by

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S5 CSE A

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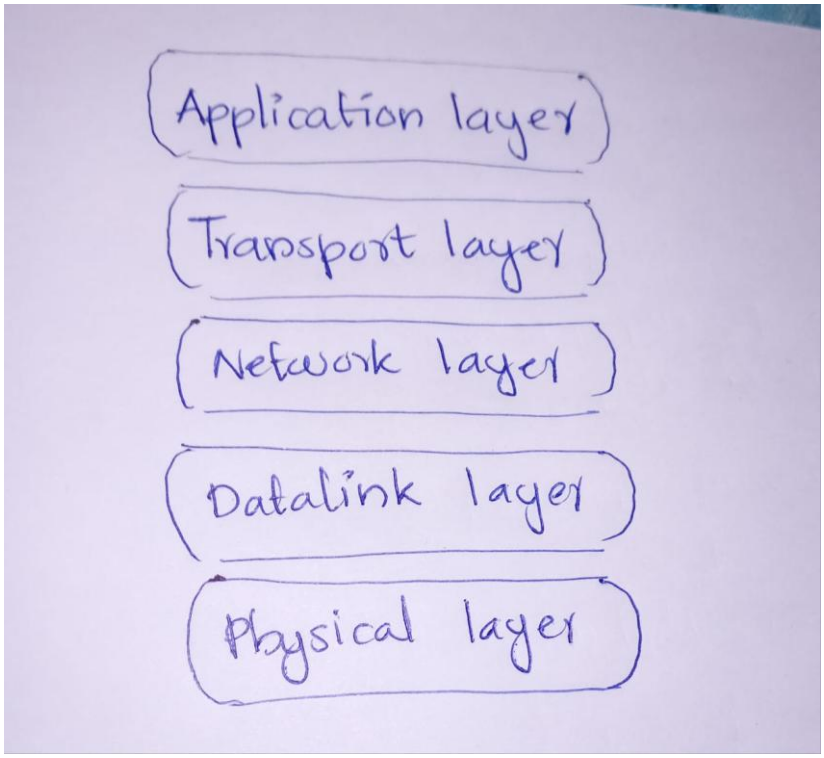
How the Internet Delivers a Web Page

Choose one website that you frequently use (e.g., Google, Amazon, YouTube, Gmail, Instagram, or your college website). Explain the complete process that occurs when a user types the website URL into a web browser and presses Enter.

Website Chosen: Google (https://www.google.com)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

   The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. Each layer performs a specific function and provides services to the layer above it.



Application Layer

The logical connection between the two application layers is end-to-end. Communication at the application layer is between two processes (two programs running at this layer). To communicate, a process sends a request to the other process and receives a response. Process-to-process communication is the duty of the application layer.

Transport Layer

The logical connection at the transport layer is also end-to-end. The transport layer is responsible for giving services to the application layer. Transmission Control Protocol (TCP) is a connection-oriented protocol that first establishes a logical connection before transferring data. User Datagram Protocol (UDP) is a connectionless protocol that transmits user datagrams without first creating a logical connection.

Network Layer

The network layer is responsible for creating a connection between the source computer and the destination computer. The communication at the network layer is host-to-host. The routers in the path are responsible for choosing the best route for each packet. The Internet Protocol (IP) is responsible for routing a packet from its source to its destination.

Data-link Layer

The data-link layer is responsible for taking the datagram and moving it across the link. TCP/IP does not define any specific protocol for the data-link layer. The data-link layer takes a datagram and encapsulates it in a packet called a frame. Some link-layer protocols provide complete error detection and correction.

Physical Layer

The physical layer is responsible for carrying individual bits in a frame across the link. Two devices are connected by a transmission medium (cable or air). The transmission medium carries electrical or optical signals. The bits received from the data-link layer are transformed and sent through the transmission media.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

Role of DNS in Converting the Domain Name into an IP Address (www.google.com)

When a user types **www.google.com** into a web browser, the browser must first find the website's IP address before it can connect. It sends a request to **a** DNS (Domain Name System) server to look up the IP address associated with the domain name. The DNS server searches its records and returns the corresponding IP address to the browser. Using this IP address, the browser establishes a connection with Google's web server and requests the webpage. DNS works like an Internet directory that converts easy-to-remember domain names into numerical IP addresses. This process allows users to access websites without having to memorize their IP addresses.

3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (Hyper Text Transfer Protocol) is the application-layer protocol that enables communication between a web browser and a web server. When a user enters a website URL, the browser creates an HTTP request and sends it to the server.

The HTTP request begins with a request line, which contains the method**,** URL, and HTTP version**.** The GET method is commonly used to request a webpage from the server. The request may also include header fields that provide additional information such as the browser type, accepted data formats, language preferences, and cookies. For methods such as POST and PUT, a message body can also be included to send data to the server.

After processing the request, the server sends an HTTP response. The response starts with a status line containing the HTTP version, a status code, and a status phrase that indicates whether the request was successful. The response also includes header fields and, if required, a message body containing the requested webpage. Finally, the browser receives the response, interprets the HTML and other resources, and displays the webpage to the user.

4. Explain How TCP Provides Reliable Communication for HTTP

TCP ensures that the webpage is delivered accurately and completely from the server to the client.

- **Connection Establishment:** Before data transfer begins, TCP creates a logical connection between the client and the server through a handshaking process. This allows both devices to prepare for communication.
- **Parameter Exchange:** During the connection setup, TCP exchanges important parameters such as packet size and buffer size to ensure smooth and efficient data transmission.
- **Reliable Data Transfer:** TCP numbers the transmitted bytes so that the receiver can detect if any data is missing or corrupted. If an error occurs, the lost data is sent again.
- **Flow Control:** TCP controls the rate of data transmission to prevent the sender from sending data faster than the receiver can handle.
- **Congestion Control:** TCP manages network traffic by adjusting the transmission speed during periods of heavy congestion, helping to maintain reliable communication.

5. Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

Google follows the **Client–Server architecture** rather than the Peer-to-Peer (P2P) model. In a client–server system, the web browser acts as the client and sends requests to Google's web server. The server processes these requests and returns the required web pages to the user.

Unlike the Peer-to-Peer model, where each computer can both provide and receive services without depending on a dedicated server, Google relies on powerful servers that continuously provide services to millions of users. Since all requests are handled by Google's servers, it is an example of the Client–Server architecture.

6. Compare HTTP and FTP

HTTP	FTP
Mainly used for accessing websites and web pages.	Mainly used for transferring files between computers.
Commonly used by web browsers.	Requires an FTP client or compatible software.
Transfers web resources such as HTML, CSS, images and videos.	Transfers files and folders of different formats.
Designed mainly for web communication.	Designed specifically for uploading and downloading files.

7. Explain How Cookies Improve User Experience with One Real-World Example

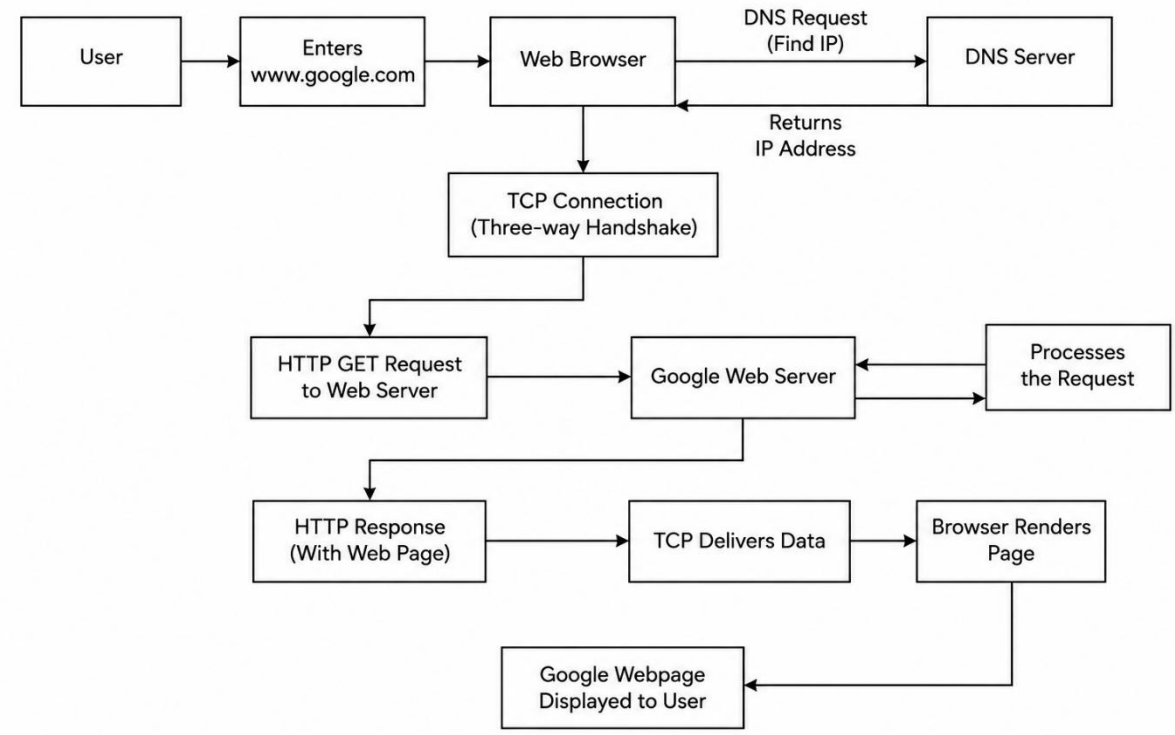
Cookies are small pieces of information created by a web server and stored by the user's browser. They help websites remember information about users, making future visits faster and more convenient.

When a browser sends a request to a website, the server creates a cookie containing user-related information and includes it in the response. The browser saves this cookie and automatically sends it back to the same server during future visits. This allows the server to recognize returning users without asking them to provide the same information again.

Example:

When a user signs in to Gmail and chooses "Remember Me", the browser stores a cookie. During the next visit, Gmail identifies the user through that cookie and automatically keeps the user logged in. This avoids entering the username and password every time, making the browsing experience faster and more convenient.

8. Flow Diagram Showing the Complete Communication



9.Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols play an important role in our daily Internet usage. Whenever users browse websites, send emails, transfer files, or communicate online, protocols such as HTTP, FTP, DNS, and SMTP make these services possible. DNS converts domain names into IP addresses, HTTP enables web browsing, and TCP ensures reliable delivery of data. The client-server architecture allows users to access services from powerful servers anywhere in the world. Together, these protocols provide fast, secure, and reliable communication across the Internet. Without application-layer protocols, modern Internet services such as Google, YouTube, online shopping, banking, and social media would not function efficiently. They form the foundation of communication between users and Internet applications